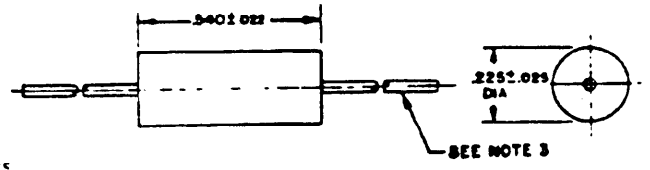
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, CERAMIC DIELECTRIC (TEMPERATURE COMPENSATING) (INSULATED), NON-ESTABLISHED RELIABILITY, STYLE CC21

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS SPECIFICATION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-20.



INCHES	MM
.022	.56
.025	.64
.225	5.72
.540	13.72

NOTES
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Leads shall be of tinned wire, .032±.002(.81±.05 mm) diameter and 1.250(31.75 mm) minimum length.

FIGURE 1. Style CC21 capacitors (insulated).

REQUIREMENTS:

Design and construction:
Dimensions and configuration: See figure 1.
Lead type: Axial.
DC rated voltage: 500 volts.
Operating temperature range: -55° to +125°C.
Characteristic: See table I.

Capacitance: Within tolerance specified (see table I).

Quality factor (Q): In accordance with MIL-C-20.

Dielectric withstanding voltage: In accordance with MIL-C-20.
Body insulation - Test 1.

Temperature coefficient and capacitance drift: In accordance with MIL-C-20.
For capacitors with values of 20 pF or less, the temperature-coefficient tolerance shall be determined in accordance with the curve shown on figure 2. For points below the curve, characteristics are not considered practicably measurable.

Life: In accordance with MIL-C-20, operating condition 1.

Marking:
Method I - In accordance with MIL-C-20, except for "JAN" brand. The beginning of the type designation shall be adjacent to the end having the inner-electrode terminal.
Method II - In accordance with MIL-C-20 and as shown on figure 3.

FSC 5910

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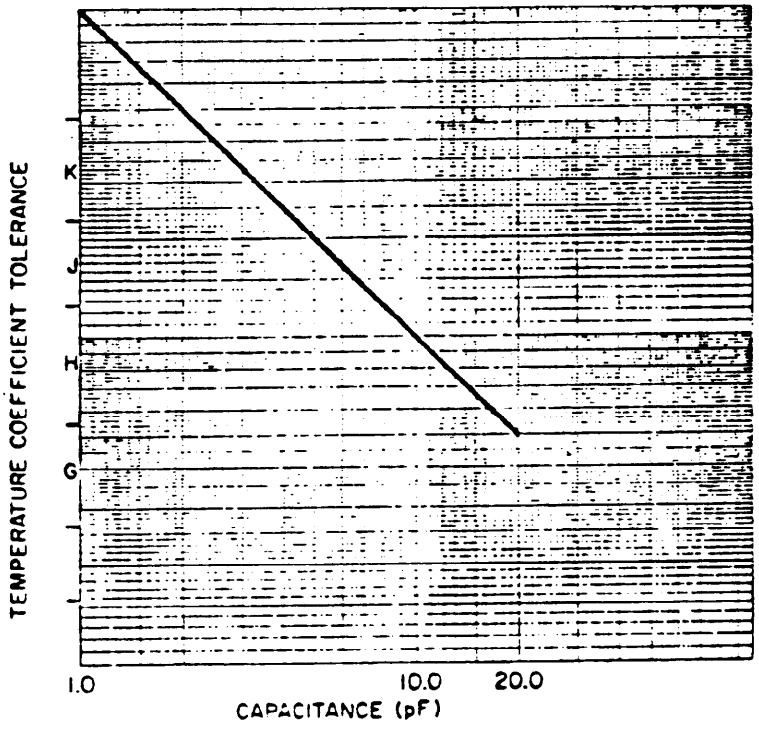
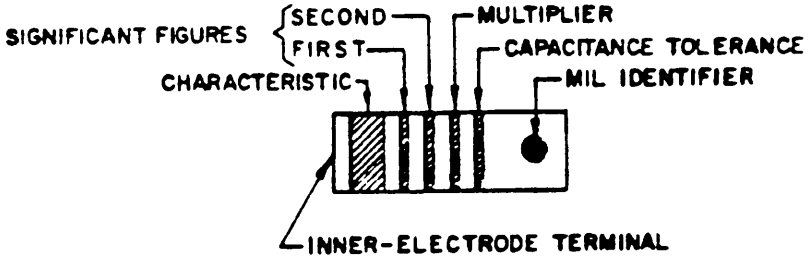


FIGURE 2. Limits for characteristic measurement.



1. The characteristic, capacitance, and capacitance tolerance shall be indicated on the capacitor by five bands of color. The characteristic band shall be adjacent to the inner-electrode terminal; the width shall be at least twice that of any other band, or the space from the characteristic band to the adjacent end of the capacitor body shall be no larger than one-third the space from the capacitance-tolerance band to the opposite end of the capacitor body.

2. The MIL identifier (black spot) shall be applied to the capacitor body adjacent to the capacitance-tolerance band.

FIGURE 3. Color-code marking.

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TABLE I. Capacitor characteristics.

Type designation 1/	Capacitance (pF)	Capacitance tolerance	Temperature coefficient shown as column head and tolerance envelope for temperature coefficient shown in column								
			A	C	H	I	P	R	S	T	U
CC21-K0R5C	0.5	C	K	K	K	K	K	K	K	K	
CC21-KR75-	0.75	CD	K	K	K	K	K	K	K	K	K
CC21-K1R0-	1	CD	K	K	K	K	K	K	K	K	K
CC21-K1R5-	1.5	CD	K	K	K	K	K	K	K	K	K
CC21-K2R0-	2	CD	K	K	K	K	K	K	K	K	K
CC21-K2R2-	2.2	CD	K	K	K	K	K	K	K	K	K
CC21-J3R0-	3	CD	J	J	J	J	J	J	J	J	J
CC21-J3R3-	3.3	CD	J	J	J	J	J	J	J	J	J
CC21—4R0-	4	CD	H	H	H	H	H	H	H	J	J
CC21—4R7-	4.7	CD	H	H	H	H	H	H	H	J	J
CC21—5R0-	5	CDF	H	H	H	H	H	H	H	J	J
CC21—6R0-	6	CDF	H	H	H	H	H	H	H	J	J
CC21—6R8-	6.8	CDF	H	H	H	H	H	H	H	J	J
CC21—7R0-	7	CDF	H	H	H	H	H	H	H	J	J
CC21—8R0-	8	CDF	H	H	H	H	H	H	H	J	J
CC21—9R0-	9	CDF	H	H	H	H	H	H	H	J	J
CC21—100-	10	CDFG	H	H	H	H	H	H	H	J	J
CC21—120-	12	GJK	H	H	H	H	H	H	H	J	J
CC21—150-	15	GJK		H	H	H	H	H	H	J	J
CC21—180-	18	GJK		H	H	H	H	H	H	J	J
CC21—200-	20	GJ		H	H	H	H	H	H	H	J
CC21—220-	22	GJK		H	H	H	H	H	H	H	J
CC21—240-	24	GJ		H	H	H	H	H	H	H	J
CC21—270-	27	FGJK				H	H	H	H	H	J
CC21—300-	30	FGJ					H	H	H	H	J
CC21—330-	33	FGJK						H	H	H	J
CC21—360-	36	FGJ						H	H	H	J
CC21—390-	39	FGJK							H	H	J
CC21—430-	43	FGJ								H	J
CC21UJ470-	47	FGJK									J
CC21UJ510-	51	FGJ									J
CC21UJ560-	56	FGJK									J
CC21UJ620-	62	FGJ									J

1/ Complete type designation will include additional letter symbols to indicate, where applicable, temperature coefficient, tolerance envelope for temperature coefficient, and capacitance tolerance.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army -
Navy - OS
Air Force - 80
DSA - ES

User activities:
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Navy - AS, SH, MC
Air Force -

Preparing activity:
Air Force - 11

Agent:
DSA - ES

(Project 5910-1064-2)

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